

$$F = G \frac{m_1 m_2}{d^2}$$

$$\phi(x) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$i\hbar \frac{\partial}{\partial t} \psi = \hat{H} \psi$$

$$F - E + V = 2$$

Economics for Engineers

Module 1

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

$$\frac{df}{dt} = \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h}$$

$$E = mc^2$$

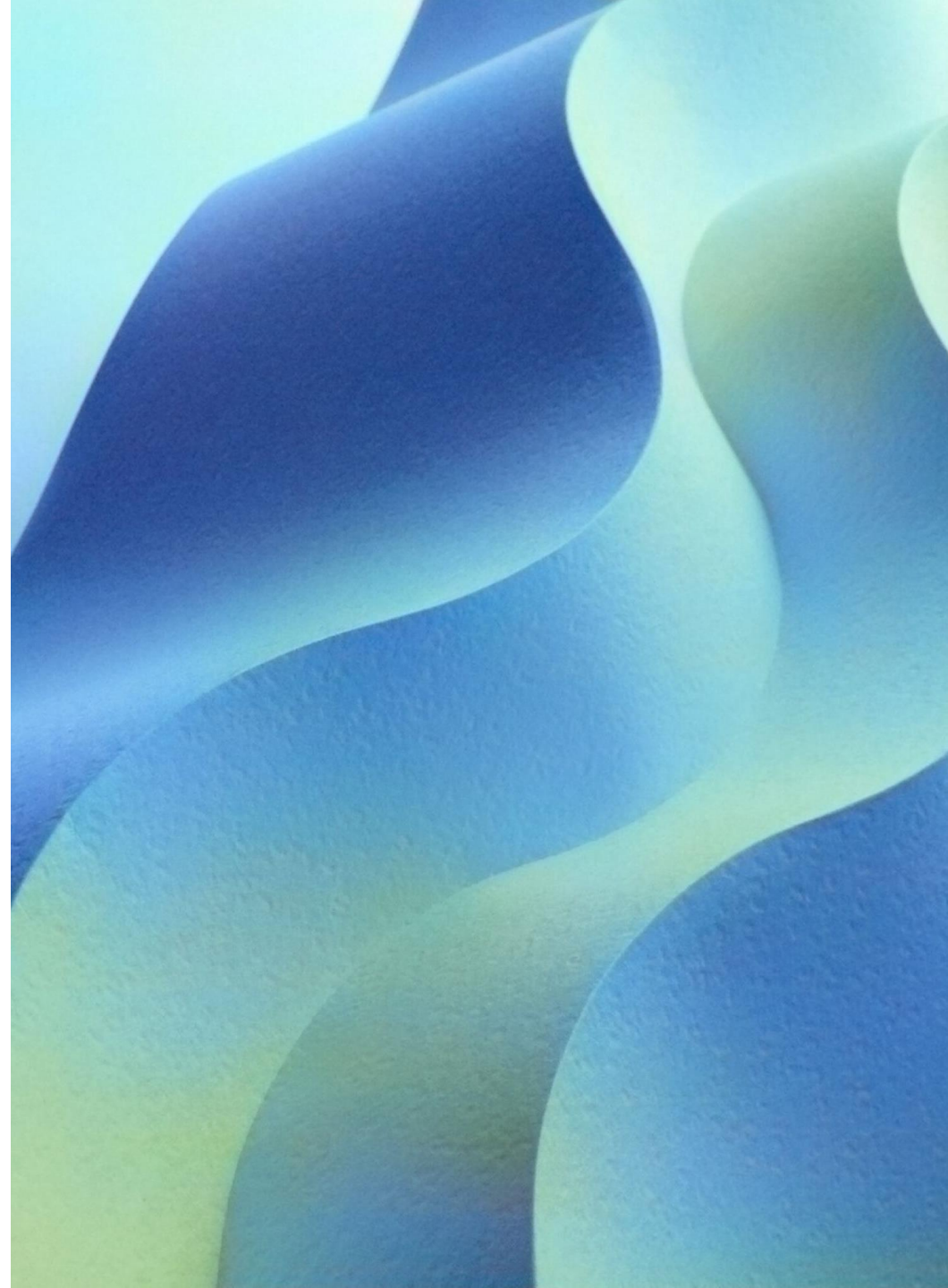
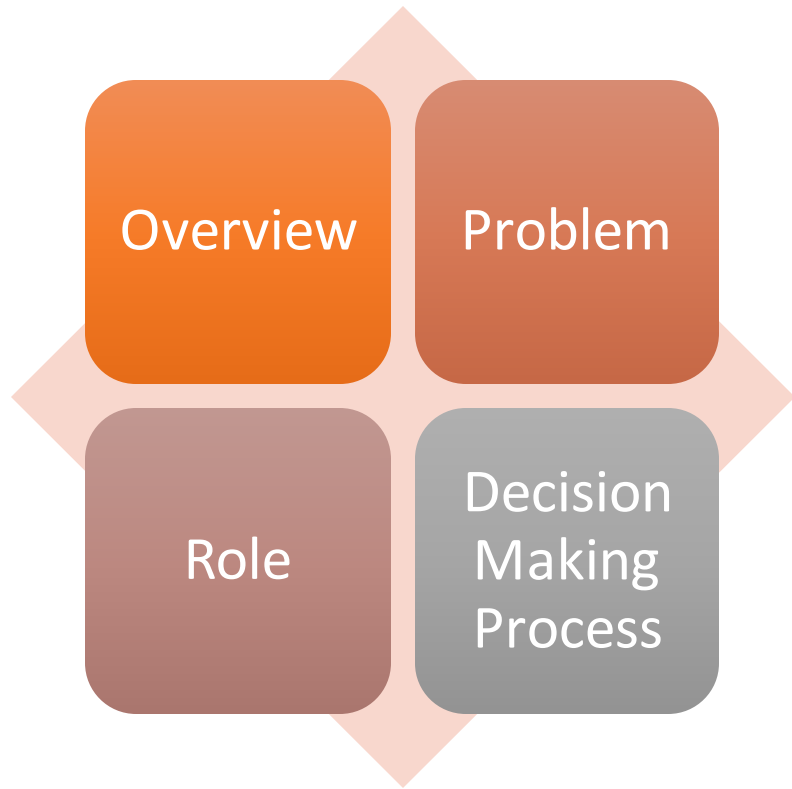
$$dS \geq 0$$

INDEX

Economic
Decision
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Engineering
Costs &
Estimation

Economic Decision Making



Engineering Costs

Fixed cost

Variable cost

Marginal & Average Costs

Sunk Costs

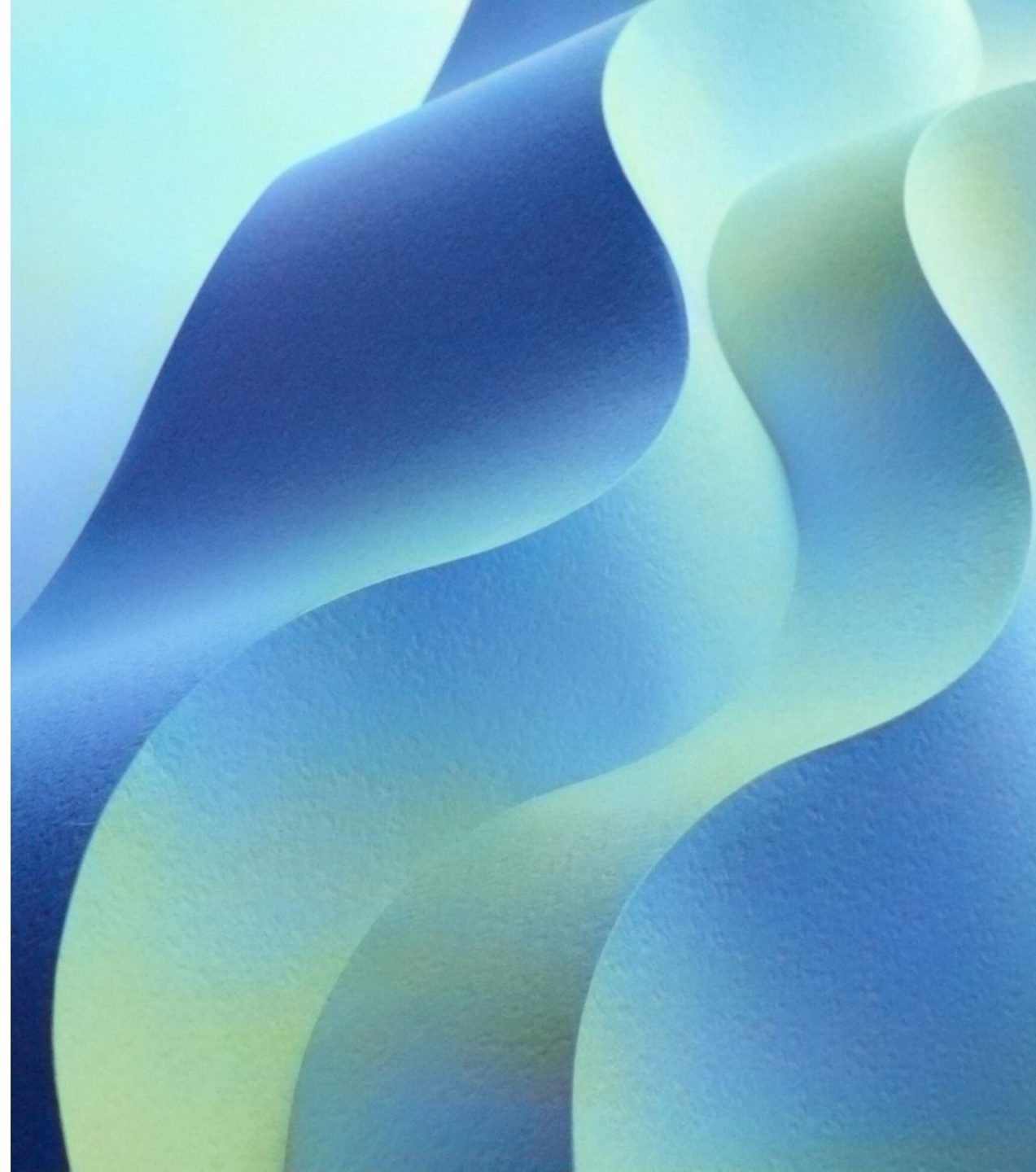
Opportunity Costs

Recurring And Nonrecurring Costs

Incremental Costs

Cash Costs vs Book Costs

Life-Cycle Costs



Engineering Estimation



TYPES OF ESTIMATE



ESTIMATING
MODELS -



PER UNIT MODEL



SEGMENTING
MODEL



COST INDEXES



POWER-SIZING
MODEL



IMPROVEMENT &
LEARNING CURVE,
BENEFITS.

Economic Decision Making



What is Economic?



Economics is the social science that studies the production, distribution, and consumption of goods and services.



What is Decision Making?



the action or process of making important decisions.

Economic Decision Making



City of
Dholakpur



A close-up photograph of a small, vibrant green seedling with two leaves sprouting from a crack in parched, greyish-brown soil. The soil is heavily cracked into irregular polygonal shapes, indicating extreme dryness. The background is a blurred continuation of the cracked earth, with some areas catching the light and others in shadow.

Define Problem
What should we develop on the land?

Clarify Goals & Priorities



Maximize use



Environmentally Sustainable



Long term positive impact



Minimal maintenance Cost



'Wow' Factor

Possible Alternatives



PARK &
SCHOOL



RETAIL STORES



HOUSING

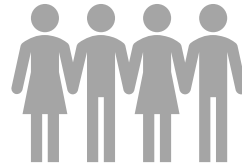


CORPORATE
DEVELOPMENT

List criteria



Cost



Community Need



Environmental impact

Weight Creation



COMMUNITY NEED – 3



ENVIRONMENTAL
IMPACT – 2



COST - 1

Evaluation

Alternative	Community need	Environmental impact	cost
Park & School	9	8	4
Retail	6	5	7
Corporate Development	5	5	9
Housing	4	6	8

Weighted Evaluation

Alternative	Community need	Environmental impact	cost
Park & School	$9 \times 3 = 27$	$8 \times 2 = 16$	$4 \times 1 = 4$
Retail	$6 \times 3 = 18$	$5 \times 2 = 10$	$7 \times 1 = 7$
Corporate Development	$5 \times 3 = 15$	$5 \times 2 = 10$	$9 \times 1 = 9$
Housing	$4 \times 3 = 12$	$6 \times 2 = 12$	$8 \times 1 = 8$

An aerial photograph of a suburban neighborhood. The image shows a grid of streets with houses, trees, and some swimming pools. The houses are mostly single-story with light-colored roofs. The trees are green and scattered throughout the neighborhood. The streets are paved and have some cars parked along the sides. The overall scene is a typical suburban residential area.

Possible alternatives

- Park & School - 47
- Retail stores - 35
- Housing - 32
- Corporate Development - 34



Make Decision

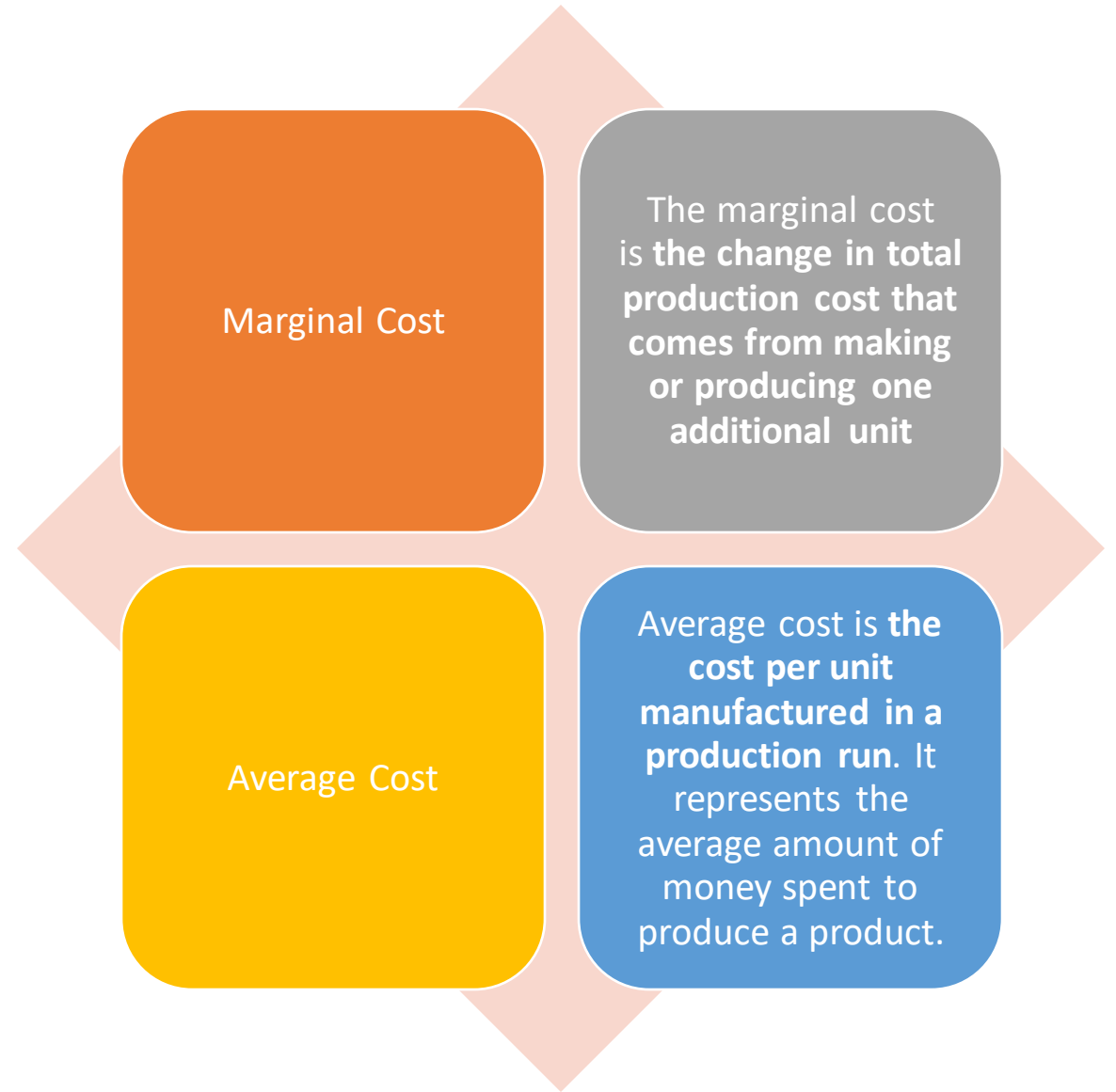
Asses Effectiveness

Engineering Costs

Engineering Costs

- Fixed Cost
- A fixed cost is **a cost that does not change with an increase or decrease in the amount of goods or services produced or sold.**
- Variable Cost
- A variable cost is **a corporate expense that changes in proportion to how much a company produces or sells**

Engineering Costs



Engineering Costs

Sunk cost

A sunk cost is **money that has already been spent and cannot be recovered.**

Opportunity cost

Opportunity cost is **the potential forgone profit from a missed opportunity—the result of choosing one alternative and forgoing another.**

Engineering Costs

Recurring Cost

Recurrent Cost is **the regular expenditure cost that is repeatedly occur for the similar goods or services on a continuing basis.**

Non recurring cost

A nonrecurring cost is **a expense that is considered a one-time or rare expense and is unlikely to happen again.**

Incremental Cost

Incremental cost is **the total cost incurred due to an additional unit of product being produced**

Engineering Costs



Cash Cost



Cash Costs are the costs Which are paid by the business while using cash or cheque, but not credit.



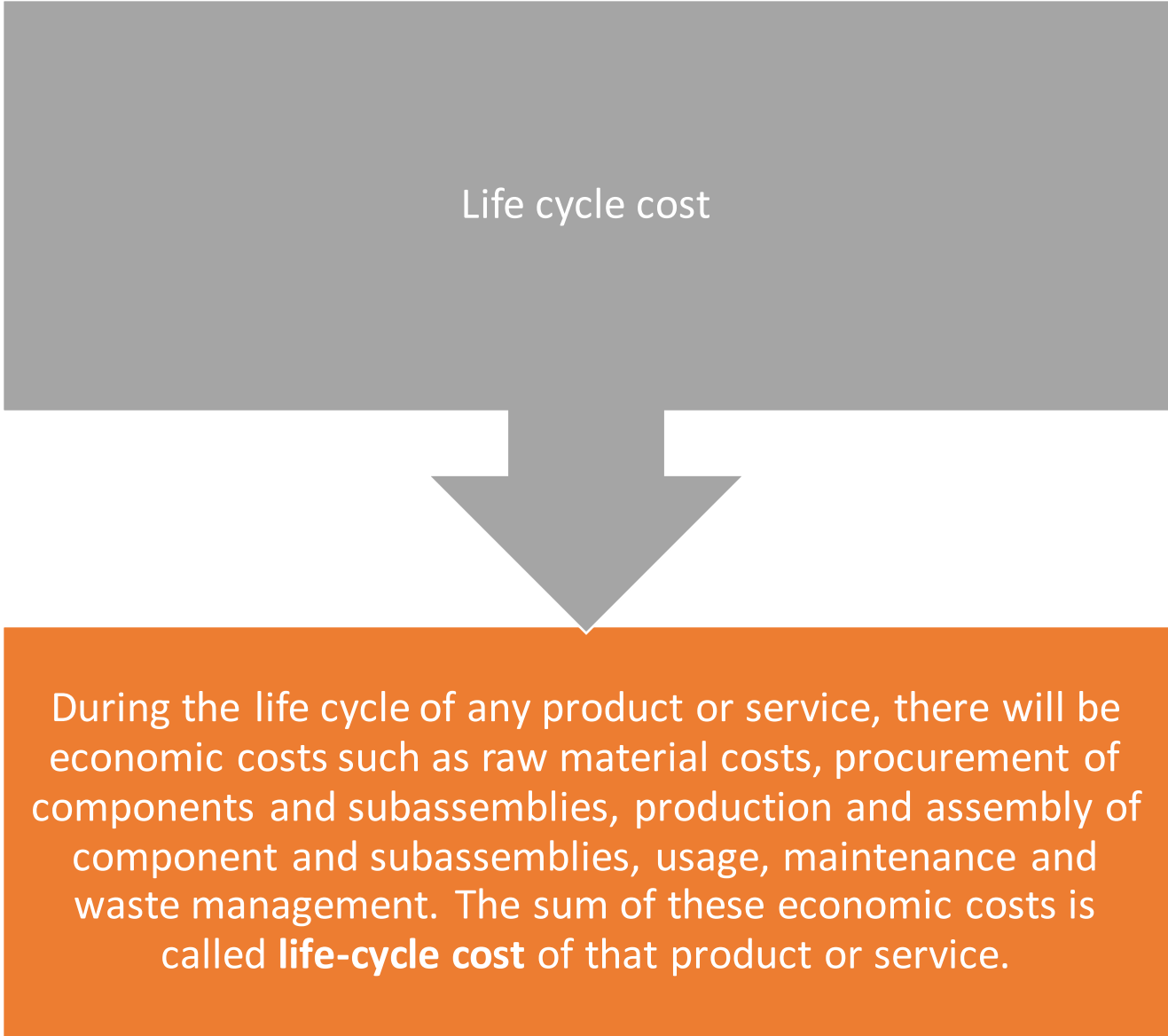
Book Cost



In any business, there are purchases of equipment and other facilities. The total money spent in buying that equipment and other facilities is called Book cost.

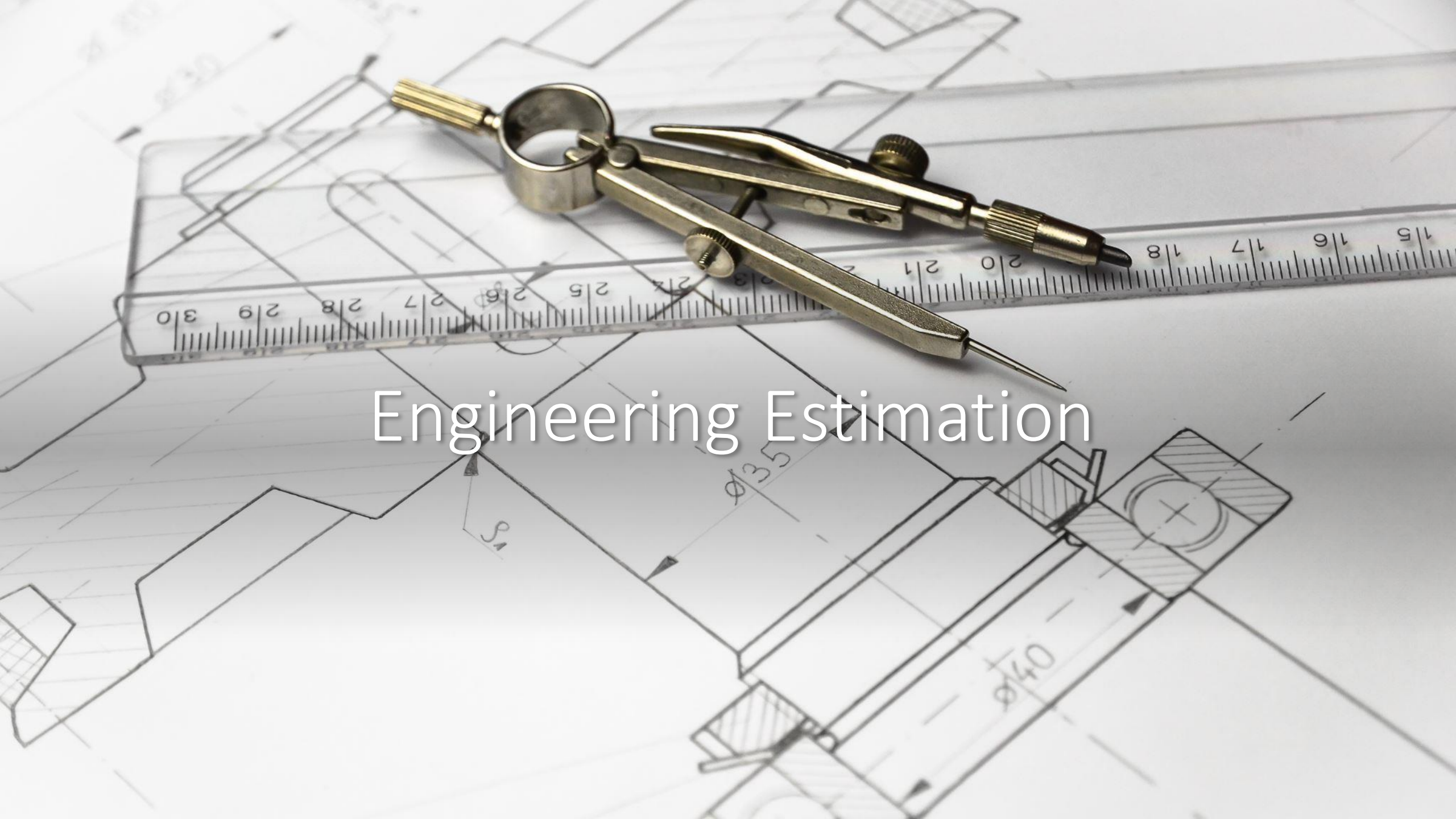
Engineering Costs

Life cycle cost



```
graph TD; A[Life cycle cost] --> B[During the life cycle of any product or service, there will be economic costs such as raw material costs, procurement of components and subassemblies, production and assembly of component and subassemblies, usage, maintenance and waste management. The sum of these economic costs is called life-cycle cost of that product or service.]
```

During the life cycle of any product or service, there will be economic costs such as raw material costs, procurement of components and subassemblies, production and assembly of component and subassemblies, usage, maintenance and waste management. The sum of these economic costs is called **life-cycle cost** of that product or service.

A detailed technical drawing of a mechanical part, possibly a shaft or housing, is shown on a blueprint. The drawing includes various dimensions and features, such as a central hole with a diameter of $\phi 35$ and a larger section with a diameter of $\phi 40$. A ruler with a scale from 15 to 30 is placed horizontally across the drawing. A pair of compasses and a pair of dividers are also visible, resting on the blueprint. The text "Engineering Estimation" is overlaid in the center of the image.

Engineering Estimation

Engineering Estimation

In a Manufacturing/service organization, estimating the cost of product/service requires utmost accuracy so that deviation between the actual cost and the estimated cost is minimal.

Engineering Estimation helps us in doing the estimation.

Per Unit Model

The per-unit model of cost estimates the cost of producing and selling one unit.

Segmentation Model

- The segmentation model of cost estimation splits the task of the product/service into different segments. Then the cost of each segment is to be estimated and finally, the cost of the product is obtained by summing the costs of all the segments of product/service.



Cost index Model

- The **cost index** describes changes in prices of cost factors relative to the selected base year.



Power sizing model

- The 'power sizing model' of cost estimation is based on **economies of scale**. It is used for estimating the **cost of equipment and other industrial items**.

Learning curve

- Learning Curve is based on the concept that **the more an individual repeats a process or activity, the more adept they become at that activity**. This translates to lower input costs and higher overall output.



An aerial photograph of a long, multi-lane highway bridge spanning a body of turquoise water. The bridge has several lanes in each direction, with white lane markings. Several vehicles, including white and blue trucks, are visible traveling across the bridge. The water is a vibrant turquoise color with visible ripples. The text "Thank you" is overlaid in a white, sans-serif font in the center of the image.

Thank you